

Central Payload Hub:

$$r_{\text{hub}} = 0.10 \text{ m } (r_{\text{hub}}/R = 0.05)$$

$$m_{\text{payload}} = 1.5 \text{ g } (m_{\text{total}} = 2.0 \text{ g})$$

Rigid Instrument Interface

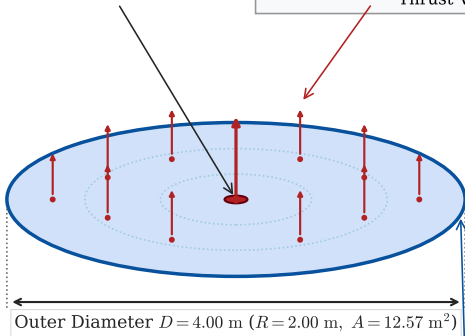
Uniform Radiation Pressure Load:

$$P_0 = 100.0 \text{ GW}, \lambda_0 = 1064 \text{ nm}$$

$$\text{Thrust Force: } F_0 = 666.40 \text{ N}$$

$$\text{Applied Load: } p_{\text{avg}} = 53.09 \text{ Pa}$$

$$\text{Thrust Vector: } +\hat{z}$$



$z (+\text{thrust})$



Clamped Tensioning Rim Boundary:

$$\text{Outer Radius: } R = 2.00 \text{ m}$$

$$\text{Displacements: } u_x = u_y = u_z = 0$$

$$\text{Rotations: } \theta_x = \theta_y = \theta_z = 0$$

$$\text{Composite: } \text{Si}_3\text{N}_4 + \text{TiO}_2/\text{SiO}_2$$